

p -adic Linear Forms in Logarithms and Universal Constraints on Wall–Sun–Sun Primes in Generalized Fibonacci Sequences

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1 Introduction

The Wall–Sun–Sun prime begins with the famous Fibonacci sequence, defined as

$$F_0 = 0, F_1 = 1, \quad F_n = F_{n-1} + F_{n-2}.$$

The **Pisano period** is defined as the period of the Fibonacci sequence mod m , given by

$$\pi(m) = \text{the period of the sequence } F_0, F_1, F_2 \dots \pmod{m}.$$

The Pisano period was first studied by Joseph-Louis Lagrange in the 18th century, and Édouard Lucas was the first to explicitly study the modular properties of Fibonacci and Lucas sequences. Then, the study of Pisano periods was made systematic in 1960 with D.D. Wall's paper *Fibonacci series modulo m* [1]. Wall's paper remarks that (denoting $k(p) = \pi(p)$):

The most perplexing problem we have met in this study concerns the hypothesis $k(p^2) \neq k(p)$. We have run a test on a digital computer which shows that $k(p^2) \neq k(p)$ for all p up to 10,000; however, we cannot yet prove that $k(p^2) = k(p)$ is impossible. The question is closely related to another one, “can a number x have the same order mod p and mod p^2 ?” for which rare cases give an affirmative answer (e.g., $x = 3, p = 11$; $x = 2, p = 1093$); hence, one might conjecture that equality may hold for some exceptional p .

Conjecturally, $\pi(p^k) = p^{k-1}\pi(p)$ [1, 2], which would imply $\pi(p^2) \neq \pi(p)$. However, it is unknown whether $\pi(p^k) = p^{k-1}\pi(p)$ for all p, k , and such counterexamples may exist. A counterexample prime that satisfies $\pi(p^2) = \pi(p)$ is called a **Wall–Sun–Sun** prime. The main significance of these primes was not discovered until 1991, with Zhi-Hong Sun and Zhi-Wei Sun's paper *Fibonacci numbers and Fermat's last theorem* [3]. In this paper, Sun and Sun discovered:

If a counterexample to Fermat's last theorem existed:

$$x^p + y^p = z^p \quad \text{for } x, y, z \in \mathbb{Z}, p \nmid xyz$$

Then p must be a Wall–Sun–Sun prime.

As such, these primes are named after D.D. Wall, Sun, and Sun. Their existence before Wiles' proof of Fermat's Last Theorem was of great importance, given the general interest in the theorem. We denote a **modified Wall–Sun–Sun–Prime** as a prime satisfying the definition of a Wall–Sun–Sun prime, $p^2 \mid F_{p-(\frac{p}{5})}$ with $(\frac{p}{5})$ as the Legendre symbol, but for a different **Lucas universe**, defined

$$U_n(P, Q) : U_n = PU_{n-1} - QU_{n-2}.$$

Then, naturally, the Wall–Sun–Sun prime extends through $p^2 \mid U_{p-(p|D)}$ with $D = P^2 - 4Q$. The main object of this paper, however, is the **extended modified Wall–Sun–Sun prime**, or a prime with $k \geq 3$ such that $p^k \mid U_{p-(p|D)}$. In particular, we bound k using three main tools.

Modular arithmetic allows us to convert the divisibility constraint of $p^2 \mid U_{p-(p|D)}$ into a single identity for any Lucas universe: $\phi_U^{2(p-(p|D))} Q^{-(p-(p|D))} - 1 \equiv 0 \pmod{p^2}$. Then, we use p -adic numbers to extend modular congruence into a p -adic valuation problem, particularly that $|\phi_U^{2(p-(p|D))} Q^{-(p-(p|D))} - 1|_p = 0$. We then use the **Bugeaud–Laurent bound** [4] on linear forms in exactly 2 p -adic logarithms to compute a bound on k .

2 Research Question

If k is taken to be the highest integer power such that $p^k \mid U_{p-(p|D)}$, then is there any nontrivial bound on k ? Is such a bound decreasing with p ? If so, is the bound strong enough that $k \rightarrow 2$ such that $p^2 \nmid U_{p-(p|D)}$ for large enough primes?

3 Methodology

We begin with the framework that p is an odd prime not equal to D , and $p^2 \mid U_{p-(p|D)}$. Assuming that p is a modified Wall–Sun–Sun prime does not lose generality since if $p^k \mid U_{p-(p|D)}$ for $k = 1$ then k is automatically bounded, and if $k \geq 3$ then we also have $p^2 \mid U_{p-(p|D)}$.

We then transition to modular arithmetic, with $U_{p-(p|D)} \equiv 0 \pmod{p^2}$. Applying **Binet’s formula** [2], given for Lucas sequences by

$$U_n = \frac{\phi_U^n - \varphi_U^n}{\sqrt{D}}, \quad \phi_U = \frac{P + \sqrt{P^2 - 4Q}}{2}, \quad \varphi_U = \frac{P - \sqrt{P^2 - 4Q}}{2}$$

we have $\frac{\phi_U^{p-(p|D)} - \varphi_U^{p-(p|D)}}{\sqrt{D}} \equiv 0 \pmod{p^2}$, and here we need the assumption $p \neq D$ such that multiplying by $\sqrt{D}$ keeps modular invertibility. We proceed in simplifying this congruence until we end up with $\phi_U^{2(p-(p|D))} Q^{-(p-(p|D))} - 1 \equiv 0 \pmod{p^2}$. From here, we apply the bound in [4] in order to bound the p -adic valuation of $\phi_U^{2(p-(p|D))} Q^{-(p-(p|D))} - 1$, which essentially becomes a bound on k through Bugeaud–Laurent.

4 Results

Lemma 1. *Let p be a modified Wall–Sun–Sun prime and $n = p - (p \mid D)$, with $(p \mid D)$ as the Legendre symbol [5]. Let $\phi_U = \frac{P + \sqrt{P^2 - 4Q}}{2}$, $\varphi_U = \frac{P - \sqrt{P^2 - 4Q}}{2}$, and $D = P^2 - 4Q$. Then we obtain*

$$\phi_U^{2n} Q^{-n} - 1 \equiv 0 \pmod{p^2}$$

which we refer to as the master congruence.

Proof. Begin with Binet’s extended formula $\phi_U = \frac{P + \sqrt{P^2 - 4Q}}{2}$, $\varphi_U = \frac{P - \sqrt{P^2 - 4Q}}{2}$, giving $U_n = \frac{\phi_U^n - \varphi_U^n}{\sqrt{P^2 - 4Q}}$ and substitute

$$\frac{\phi_U^{p-(p|D)} - \varphi_U^{p-(p|D)}}{\sqrt{D}} \equiv 0 \pmod{p^2}, \quad p \neq 2, D.$$

Assuming $p \nmid D$ and working in the quadratic field $\mathbb{Q}(\sqrt{D})$ we have that $\sqrt{D}$ is a unit [6] and therefore can multiply by $\sqrt{D}$ retaining invertibility, giving $\phi_U^{p-(p|D)} - \varphi_U^{p-(p|D)} \equiv 0 \pmod{p^2}$. Divide by $\varphi_U^{p-(p|D)}$, which is invertible since $p \nmid Q$ and $\phi_U \varphi_U = Q$, so both ϕ_U and φ_U are p -adic units, to obtain

$$\left(\frac{\phi_U}{\varphi_U}\right)^{p-(p|D)} - 1 \equiv 0 \pmod{p^2}.$$

Since $\phi_U/\varphi_U = \frac{\phi_U^2}{Q}$ from Vieta's formulas [7, 5], we have the Master Congruence

$$\left(\frac{\phi_U^2}{Q}\right)^{p-(p|D)} - 1 \equiv 0 \pmod{p^2} \implies \phi_U^{2n} Q^{-n} - 1 \equiv 0 \pmod{p^2}.$$

□

Theorem 1 (Bugeaud–Laurent, p -adic Two–Logarithm Estimate). *Let p be an odd prime. Let K be a number field containing $a_1, a_2 \in K^\times$, and let $d = [K : \mathbb{Q}]$. Fix an extension of the p -adic valuation v_p to K . Assume that*

1. a_1 and a_2 are p -adic units (i.e. $v_p(a_i) = 0$ for $i = 1, 2$);
2. b_1, b_2 are nonzero integers with $\gcd(p, b_1, b_2) = 1$ (note this does not require $\gcd(b_1, b_2) = 1$);
3. $L := a_1^{b_1} a_2^{b_2} - 1 \neq 0$.

For $i = 1, 2$ define the (logarithmic) absolute Weil height $h(a_i)$ and choose real numbers A_i satisfying

$$\log A_i \geq \max\{h(a_i), \frac{1}{d}|\log_p(a_i)|_p, \frac{\log p}{d}\}.$$

Put

$$b_0 = \frac{|b_1|}{d \log A_2} + \frac{|b_2|}{d \log A_1}.$$

Then there exists an explicit absolute constant C_p depending only on p such that

$$v_p(L) \leq C_p d^4 (\log A_1)(\log A_2) \left(\max\{\log b_0 + \log \log p + c_0, p \log p\} \right)^2,$$

where c_0 is an explicit small absolute constant (one may take $c_0 = 0.64$ as in [4]). If a_1 and a_2 are multiplicatively independent, the same inequality holds with a (slightly) smaller explicit constant C'_p in place of C_p .

Remarks.

- The constants C_p, C'_p are explicit and arise from the numerical form of Bugeaud–Laurent, Theorems 1 and 2 [4] in the p -adic case; in applications, one may replace them by the concrete values obtained by following the numerical constants in the paper.
- The hypothesis that a_i are p -adic units and $L \neq 0$ ensures the p -adic logarithms appearing in the proof are defined on the relevant $1 + p\mathbb{Z}_p$ -neighborhood.
- No auxiliary composite- m hypotheses (H1),(H2) are required in this prime (p)-adic specialization.

Theorem 2. Let $\Xi = \phi_U^{2(p-(p|D))} Q^{-(p-(p|D))} - 1$ and $\Xi \neq 0$. Then,

$$v_p(\Xi) \leq \begin{cases} 2400p & \text{if } (p | D) = 1, \\ 2400(p^2 + p) & \text{if } (p | D) = -1 \end{cases}$$

Proof. Checking Hypotheses

We apply Theorem 1 with the parameters specified below. No optimization of constants was attempted¹. We work in the quadratic field $\mathbb{Q}(\sqrt{D})$ with the p -adic valuation v_p , and since $p \nmid D$, this extension is unramified.

In our application, we work in $K = \mathbb{Q}(\sqrt{D})$ and fix an extension of the p -adic valuation v_p . We assume $p \nmid D$ and $p \nmid Q$, hence ϕ_U, φ_U, Q are p -adic units (since $\phi_U \varphi_U = Q$); thus the p -adic logarithms appearing below are defined on the $1 + p\mathbb{Z}_p$ -neighborhood. Moreover, with $n = p - (p | D) \in \{p - 1, p + 1\}$ we have $p \nmid n$, so $\gcd(p, 2n, -n) = 1$, which is the coprimality condition required in the p -adic theorem. Finally, we assume $\Xi = \phi_U^{2n} Q^{-n} - 1 \neq 0$; the (exceptional) case $\Xi = 0$ is algebraically detectable and may be treated separately.

From this point onward we assume $p \geq p_0(P, Q)$ so that the terms $\frac{\log p}{d}$ dominate the fixed height contributions $h(\phi_U), h(Q)$ and the p -adic logarithmic terms; any finite set of small primes $p < p_0$ can be checked directly or bounded numerically. Under this convention, the simplifications below are valid.

¹A potential future optimization of this result is to optimize the constants used in the application of [4] in order to derive a smaller constant than 2400.

Substitutions and Simplification

Set $a_1 = \phi_U$, $a_2 = Q$, $b_1 = 2n$, and $b_2 = -n$ to have

$$v_p(\Xi) \leq \frac{24p(p^f - 1)}{(p-1)(\log(p))^4} d^4 \left(\max \left\{ \log(b') + \log(\log(p)) + 0.4, \frac{10 \log(p)}{d}, 10 \right\} \right)^2 \log(A_1) \log(A_2)$$

- $f = \frac{1}{2}(3 - (p \mid D))$

- $d = \left[\mathbb{Q} \left(\frac{P + \sqrt{P^2 - 4Q}}{2} \right) : \mathbb{Q} \right] = \begin{cases} 1 & \text{if } P^2 - 4Q \text{ is a perfect square} \\ 2 & \text{otherwise} \end{cases}$

- $h \left(\frac{P + \sqrt{P^2 - 4Q}}{2} \right) = \frac{1}{2} \left(\log \left(\max \left\{ 1, \left| \frac{P + \sqrt{P^2 - 4Q}}{2} \right| \right\} \right) + \log \left(\max \left\{ 1, \left| \frac{P - \sqrt{P^2 - 4Q}}{2} \right| \right\} \right) \right)$

This expression relies on the values of the so-called "golden ratios", the roots of $x^2 - Px + Q$. Using the logarithm product property we have $h(\phi_U) = \frac{1}{2} (\log(\max\{1, |\phi_U|\} \max\{1, |\varphi_U|\}))$. Since the roots either:

1. Are both < 1 , in which case the argument becomes $1 \cdot 1 = 1$.
2. Have both > 1 in which case simplifies to $|\phi_U \varphi_U| = |Q|$ from Vieta's formulas.
3. Have one < 1 and the other > 1 . Without loss of generality take $|\phi_U| > 1$ and $|\varphi_U| < 1$ giving the argument $1 \cdot |\phi_U| = |\phi_U|$.

Simplifying, we have $h(\phi_U) = \frac{1}{2} (\log(\max\{1, |Q|, |\phi_U|\}))$.

- $\log(A_1) = \max \left\{ h(\phi_U), \frac{|\log(\phi_U)|_p}{d}, \frac{\log(p)}{d} \right\} = \max \left\{ \frac{1}{2} \log \max\{1, |Q|, |\phi_U|\}, \frac{|\log(\phi_U)|_p}{d}, \frac{\log(p)}{d} \right\}$.

Since all but $\frac{\log(p)}{d}$ are constant and $\frac{\log(p)}{d} \rightarrow \infty$ we take for large enough p that $\log(A_i) = \frac{\log(p)}{d}$. $\log(A_2) = \max \left\{ h(Q), \frac{|\log(\phi_U)|_p}{d}, \frac{\log(p)}{d} \right\}$, and by the same argument as with $\log(A_1)$, we take $\log(A_2) = \frac{\log(p)}{d}$.

- $b' = \frac{|b_1|}{d \log(A_2)} + \frac{|b_2|}{d \log(A_1)} = \frac{2n}{\log(p)} + \frac{n}{\log(p)} = \frac{3n}{\log(p)}$.

Plugging in these values, we have:

$$v_p(\Xi) \leq \begin{cases} \frac{24p(p^{\frac{1}{2}(3-(p \mid D))} - 1)}{(p-1)(\log(p))^2} \left(\max \{ \log(3n) + 0.4, 10 \log(p), 10 \} \right)^2 & \text{if } P^2 - 4Q \text{ is a perfect square,} \\ \frac{96p(p^{\frac{1}{2}(3-(p \mid D))} - 1)}{(p-1)(\log(p))^2} \left(\max \{ \log(3n) + 0.4, 5 \log(p), 10 \} \right)^2 & \text{otherwise.} \end{cases}$$

Since $\log(3n) \approx \log(p) + \log(3) \approx \log(p) + 1$ and $10 \log(p) > \log(n) + 1 > 10$ for sufficient p , we take the maximum to be $10 \log(p)$, and therefore the expansion to be $100 \log(p)^2$, which simplifies to:

²We set $p > p_0(P, Q)$ as an explicit prime such that this constraint holds, and the finite set of $p \leq p_0(P, Q)$ is easily checkable with computation for outliers.

$$v_p(\Xi) \leq \frac{2400p(p^{\frac{1}{2}(3-(p|D))} - 1)}{(p-1)} = \begin{cases} 2400p & \text{if } (p | D) = 1 \\ 2400(p^2 + p) & \text{if } (p | D) = -1 \end{cases}$$

Computing the term for $(p | D) = -1$ we have $2400 \frac{p}{p-1} (p^2 - 1) = 2400 \frac{p(p-1)(p+1)}{(p-1)} = 2400p(p+1) = 2400(p^2 + p)$. □

To the best of our knowledge, no upper bound of $v_p(U_{p-(p|D)})$ yet exists in the literature.

The Case $\Xi = 0$

We consider the case $\phi_U^{2(p-(p|D))} Q^{-(p-(p|D))} - 1 = 0$ and have that $\left(\frac{\phi_U^2}{Q}\right)^{p-(p|D)} = 1$. We thus have that $\frac{\phi_U^2}{Q}$ must be a $p - (p | D)$ -th root of unity. Since ϕ_U^2/Q is purely real, we disregard imaginary roots. Then, recalling that $\phi_U^2/Q = \phi_U/\varphi_U$ we have $\phi_U/\varphi_U = \pm 1$. Then, we put $P + \sqrt{P^2 - 4Q} = \pm (P - \sqrt{P^2 - 4Q})$. Taking the positive case we see $\phi_U = \varphi_U$, which cannot be true unless $D = 0$, which is not a valid Lucas sequence. Examining the negative case we have $P + \sqrt{P^2 - 4Q} = -P + \sqrt{P^2 - 4Q} \implies 2P = 0$, and thus $P = 0$, which is also not a valid Lucas sequence. Therefore, $\Xi \neq 0$ for all valid Lucas sequences.

5 Discussion

5.1 Interpretation of the Main Bound

The bound tells us that $p^{2400p} \nmid U_{p-(p|D)}$ if $(p | D) = 1$ and $p^{2400p^2+2400p} \nmid U_{p-(p|D)}$ if $(p | D) = -1$ for any Lucas universe with valid coefficients such that $p \nmid D$, $D \geq 1$, and p being an odd prime. The constant being the same for every prime across every Lucas universe shows that a deep p -adic "shape" governs divisibility.

5.2 Sharpness and Quality of the Main Bound

The bound we have derived is **not sharp**. In fact, the constant is quite large as a result of the nature of Baker-type theorems, and it is very fast-growing. However, this constitutes the first bound on divisibility known to us, and presents a significant step towards potentially proving that Wall-Sun-Sun primes can/cannot exist in the Fibonacci sequence.

5.3 Comparison to a Trivial Size-Based Bound

We have that $U_n \sim \frac{\phi_U^n}{\sqrt{D}}$. Since any factor of a number must be strictly less than the number we have $p^k \lesssim \frac{\phi_U^{p-(p|D)}}{\sqrt{D}}$, and therefore $k \ln(p) \lesssim (p - (p | D)) \ln(\phi_U) - \frac{1}{2} \ln(D)$, and thus $k \lesssim n \frac{\ln(\phi_U)}{\ln(p)} - \frac{1}{2 \ln(p)} \ln(D)$. Visiting cases where the size bound is worse than the ideal bound, we have $(p - (p | D)) \frac{\ln(\phi_U)}{\ln(p)} - \frac{1}{2 \ln(p)} \ln(D) > 2400p$ and discarding the Legendre term and constant term we have $\ln(\phi_U) \frac{p}{\ln(p)} > 2400p \implies \frac{\ln(\phi_U)}{\ln(p)} > 2400$ and simplifying we have $\phi_U > p^{2400}$.

Since we allow p to be "large enough",

5.4 Dependence on the Legendre Symbol

While the Legendre symbol dependence may seem arbitrary or like an artifact of Bugeaud–Laurent, we have a simple hypothesis for why it affects the bound: when $(p \mid D) = 1$, the term is U_{p-1} , while when $(p \mid D) = -1$ the term is U_{p+1} . Since $U_{p+1} \approx \phi_U^2 U_{p-1}$, this extra size allows for higher divisibility and more p -adic flexibility. While this is not rigorously proven or supported, we conjecture that the size of U_n is critical to the number theoretic structure of extended Wall–Sun–Sun primes, and the p -adic analysis reflects this.

5.5 Comparison with Computed Results

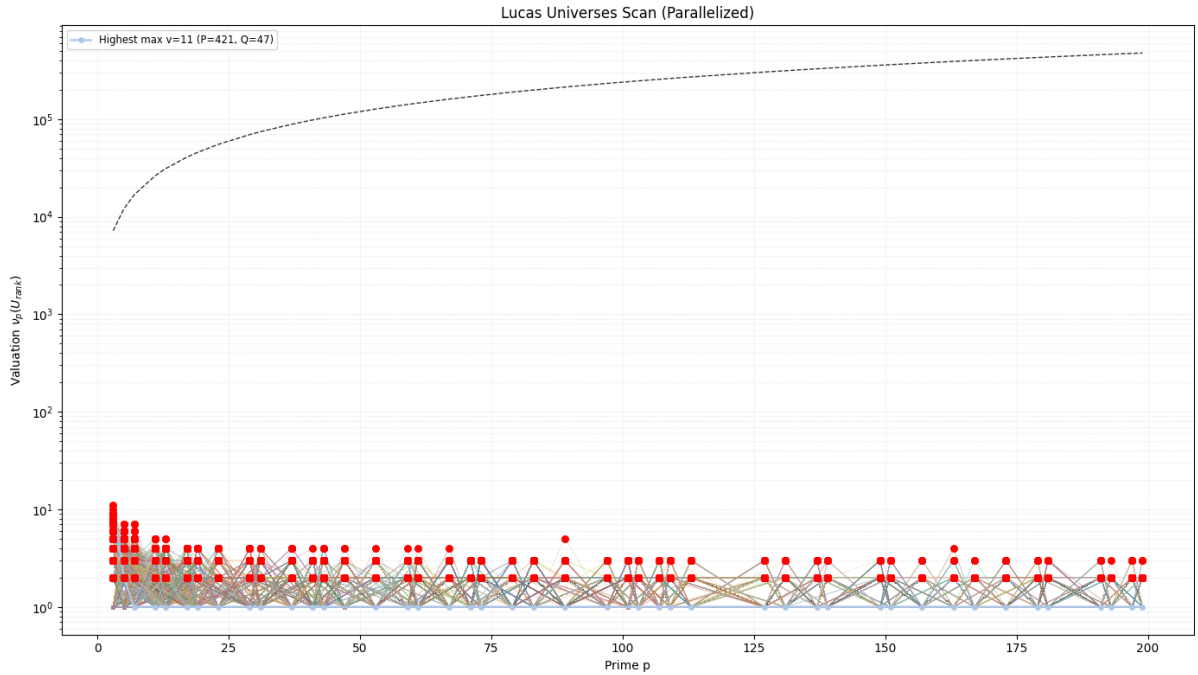


Figure 1: A graph generated with Python 3.14 [8], Numpy [9], Sympy [10], and Matplotlib [11] of divisibility points from universes with $P \in [1, 750]$ and $Q \in [-375, 375] \setminus 0$ not including degenerate universes. Iterated over a total of 562456 universes and 45 primes per universe (46 primes up to 200, and not counting primes equal to the discriminant gives the estimate of 45 per universe) for a total of $\approx 25,310,520$ data points. [12]

We skipped universes with problematic traits:

- Periodic Lucas universes contain $U_n = 0$ predictably for infinitely many n . We ignore these for computation since they are bounded by small integers, e.g. $U_n(1, 1) \in \{-1, 0, 1\}$, and so all primes eventually either do not divide at all (which is impossible for standard sequences), or divide 0 infinitely many times, defeating the purpose of a Wall–Sun–Sun prime.
- Other universes have zeroes for every $U_{k(P+Q)}$, but are not periodic. These are also ignored because $p^k \mid 0$ is meaningless.

- The final kind of degenerate universe is one with both P and Q divisible by some prime. If $P = kp$ and $Q = lp$ then we have $U_n = kpU_{n-1} - lpU_{n-2} = p(kU_{n-1} - lU_{n-2})$, and this becomes even worse if k, l are close in prime factors. Thus, these universes are trivially and highly divisible by primes, so we ignore them.

5.6 Methodological Significance

Our methodology presents further opportunities for improvement. For example, Bugeaud–Laurent [4] itself is an optimization of Yu’s theorem [13], which is in turn a p -adic analogue of Baker’s theorem [14]. Further optimization of the bound can come from either optimizing the constants used to obtain a better final constant (e.g., replacing ϕ_U^2 and Q with $\phi_U \varphi_U$) or from optimizing the theorem on linear forms in logarithms itself. This means that **this technique can be optimized to achieve better results.**

6 Limitations and Scope

The large constant presented is a known limitation of Baker’s theorem type bounds, which are known to be effective but not sharp [14]. While we apply a sharper version of Baker, such methods require large constants in order to cover every case in generality. No attempt was made to optimize the constants in the application of Bugeaud–Laurent or optimize the number of p -adic logarithms. For example, one may explore $\left(\frac{\phi_U^2}{Q}\right)^n - 1$ in one p -adic logarithm rather than two.

Throughout the paper we also assume that $p \nmid Q$ and $p \nmid D$, ensuring that certain constants in theorem 2 remain p -adic units. We also assume that p is greater than an explicit constant dependent only on P and Q , and the finite set of primes less than this constant can be checked numerically for divisibility. While our bound is not optimal, it is uniform in all Lucas universes and constitutes a restriction on the p -adic framework of Lucas sequences. In computation, we ignore all universes that present misleading or uninformative divisibility, such as $U_n = 0$ and cases where divisibility is not structural but "cheated" by P and Q .

Taken together, these limitations show the reach of current, unoptimized p -adic methods, rather than intrinsic structural limitations on Lucas sequences. The structure of the bound suggests that further improvements may be possible through alternative formulations of the master congruence or through sharper logarithmic estimates.

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